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Inventor(s)	Cohen; Noy et al.

Folded macro-tele camera lens designs including six lenses of ++-+-+ or +-++-+, seven lenses of ++-++-+, or eight lenses of ++-++-++ refractive powers

Abstract

Folded digital cameras comprising a lens system with a lens and an image sensor, the lens having $N \geq 6$ lens elements $L_{sub.i}$, an effective focal length (EFL) and a total track length (TTL), wherein each lens element has a respective focal length $f_{sub.i}$ and wherein a first lens element $L_{sub.1}$ faces an object side, and an optical path folding element (OPFE) for providing a folded optical path between an object and the lens. In some embodiments, the lens system has a focusing range that covers object-lens distances from infinity to a minimal object distance (MIOD), wherein $MIOD/EFL$ is smaller than 20 or even 7. In some embodiments, the ratio of a maximal chief ray angle to a field of view of the folded camera $Max\ CRA/FOV$ is smaller than 0.25 or even 0.15 when the camera is focused at infinity.

Inventors: Cohen; Noy (Tel Aviv, IL), Rudnick; Roy (Tel Aviv, IL), Dror; Michael (Tel Aviv, IL), Goulinski; Nadav (Tel Aviv, IL), Goldenberg; Ephraim (Tel Aviv, IL)

Applicant: Corephotonics Ltd. (Tel Aviv, IL)

Family ID: 1000008781080

Assignee: Corephontonics Ltd. (Tel Aviv, IL)

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Primary Examiner: Huang; Wen

Attorney, Agent or Firm: Nathan & Associates

Background/Summary

CROSS REFERENCE TO EXISTING APPLICATIONS (1) This is a continuation from U.S. patent application Ser. No. 18/412,545 filed Jan. 14, 2024 (now allowed), which was a continuation from U.S. patent application Ser. No. 17/614,380 filed Nov. 26, 2021 (now U.S. Pat. No. 11,914,117), which was a 371 application from international patent application PCT/IB2021/056358 filed Jul. 14, 2021, and is related to and claims priority from U.S. Provisional Patent Applications No. 63/059,200 filed Jul. 31, 2020, and 63/070,501 filed Aug. 26, 2020, which are expressly incorporated herein by reference in their entirety.

FIELD

(1) The presently disclosed subject matter is related in general to the field of digital cameras and in particular to folded lenses in such cameras.

DEFINITIONS

(2) In this application and for optical and other properties mentioned throughout the description and figures, the following symbols and abbreviations are used, all for terms known in the art: Total track length (TTL): the maximal distance, measured along an axis parallel to the optical axis of a lens, between a point of the front surface $S_{\text{sub}.1}$ of a first lens element $L_{\text{sub}.1}$ and an image sensor, when the system is focused to an infinity object distance. Back focal length (BFL): the minimal distance, measured along an axis parallel to the optical axis of a lens, between a point of the rear surface $S_{\text{sub}.2N}$ of a last lens element $L_{\text{sub}.N}$ and an image sensor, when the system is focused to an infinity object distance. Effective focal length (EFL): in a lens (assembly of lens elements $L_{\text{sub}.1}$ to $L_{\text{sub}.N}$), the distance between a rear principal point P' and a rear focal point F of the lens. f-number, $(f/\#)$: the ratio of the EFL to an entrance pupil diameter. Chief ray angle (CRA): a (image-side) chief ray is the line between the center of the exit pupil of a lens system and an image point. The CRA is the angle between a chief ray and the optical axis of the lens. Here and in the following, we refer to image-side chief rays and CRAs, in contrast to an object-side chief ray,

which is the line between an object point and the center of the entrance pupil of a lens system. A maximal CRA (Max CRA) is the largest CRA present in a lens system considering all fields supported by the lens system.

BACKGROUND

(3) Multi-aperture cameras (or “multi-cameras”, of which a “dual-cameras” having two cameras is an example) are becoming the standard choice of portable electronic mobile device (e.g. smartphones, tablets, etc.) makers. A multi-camera setup usually comprises a wide field-of-view (or “angle”) FOV.sub.W camera (“Wide” camera or “W” camera), and at least one additional camera, either with the same FOV (e.g. a depth auxiliary camera), with a narrower (than FOV.sub.W) FOV (Telephoto or “Tele” camera with FOV.sub.T), or with an ultra-wide field of view FOV.sub.UW (wider than FOV.sub.W, “UW” camera).

(4) FIGS. 1A and 1B illustrate a known digital folded camera **100**. Camera **100** comprises an optical path folding element (OPFE) **102** e.g. a prism or mirror, a lens **104** with a plurality of lens elements (not visible in this representation, but visible e.g. in FIGS. 2A-D) and an image sensor **106**. The lens elements may be (as in FIGS. 2A-2D) axial symmetric along an optical axis **108**. In other embodiments, the lens elements may not be axial symmetric. For example, lens elements may be cut (diced, sliced) into a non-circular shape (not shown).

(5) At least some of the lens elements may be included in a “barrel” **110**. The barrel may have a longitudinal symmetry along optical axis **108**. In FIGS. 1A to 1C, the cross-section of this barrel is circular. This is however not mandatory and other shapes can be used, e.g. for hosting cut lens elements.

(6) The path of the optical rays from an object (not shown) to image sensor **106** defines an optical path (see optical paths **112** and **114**, which represent portions of the optical path). OPFE folds the optical path from a first optical path **112** to a second optical path **114**, the latter being substantially parallel to optical axis **108**.

(7) In particular, in some examples, OPFE **102** is inclined at substantially 45 degrees with respect to optical axis **108**. In FIG. 1A, OPFE **102** is also inclined at substantially 45 degrees with respect to optical path **112**.

(8) In some known examples, image sensor **106** lies in a X-Y plane substantially perpendicular to optical axis **108**. This is however not limiting, and image sensor **106** can have a different orientation. For example, and as described in international published patent application WO2016/024192, image sensor **106** may lie in the XZ plane. In this case, an additional OPFE can be used to reflect the optical rays towards image sensor **106**.

(9) Two cameras, for example a Wide camera **100** and a regular UW camera **130** may be included in a digital camera **150** (also referred to as dual-camera). A possible configuration is shown in FIG. 1C. UW camera **130** may include an aperture **132** (indicating the object side of the camera) and an optical lens system **134** (or “Wide lens module”) with a symmetry (and optical) axis **136** in the Y direction, as well as a UW image sensor **138**. The UW camera comprises a UW lens system configured to provide a UW image. As already indicated, the UW camera has a field of view FOV.sub.UW larger than FOV.sub.W. For example, FOV.sub.UW may be 80-130 degrees and FOV.sub.W may be 60-90 deg. Notably, in other examples, a plurality of Wide cameras and/or a plurality of Tele cameras can be incorporated and operative in a single digital camera. The FOV.sub.T of a Tele camera may be for example 20-50 degrees.

(10) A “Macro-photography” mode is becoming a popular differentiator for smartphone cameras. “Macro-photography” refers to photographing objects close to the camera, so that an image recorded on the image sensor is nearly as large as the actual object photographed. The ratio of image size to object size is the object-to-image magnification M, defined by:

$$(11) M = -\frac{v}{u} = -\frac{\text{dist}(2^{\text{nd}}\text{principal plane, Image})}{\text{dist}(\text{Object, } 1^{\text{st}}\text{principal plane})}$$

where v is a lens-image distance defined by the distance of the 2.sup.nd (or “rear”) principal plane of the lens and the image, and u is an object-lens distance defined by the distance of the object to

the 1.sup.st (or “front”) principal plane of the lens. The minus sign is generally not mentioned explicitly.

(12) In the context of digital single-lens reflex (DSLR) cameras, a Macro image is defined by having a M of about 1:1 or larger, e.g. 1:1.1. In the context of smartphones, “Macro image”. may refer to images with M of about 10:1 or even 15:1. First smartphone models have entered the consumer market that provide Macro-photography capabilities, usually by enabling very close focusing with a UW camera, which has a relatively short EFL (e.g. 2.5 mm).

(13) A UW camera can focus to the close range required for Macro photography (e.g., 1.5 cm to 15 cm), but its spatial resolution is poor since its focal length is small and its FOV is large. For example, consider a UW camera with 2.5 mm focal length. When focused to an object at 5 cm (lens-object distance), the UW camera will have approximately M=19:1. This according to thin lens equation

$$(14) \frac{1}{EFL} = \frac{1}{u} + \frac{1}{v}$$

with EFL=2.5 mm, v=2.6 mm and u=50 mm. Even when focused to as close as 1.5 cm, the M of the UW camera will be approximately 5:1. Capturing objects in Macro images from these short object-lens distances of e.g. u=5 cm or less is very challenging for a user. For example, it may render framing of the image very difficult, it may prohibit taking images of popular Macro objects such as living subjects (e.g. insects), and it may introduce shadows and obscure the lighting in the scene. Additionally, an UW camera has a relatively large depth of field (DoF), even for Macro images. The relatively large DoF corresponds to a low degree of optical Bokeh, which is a highly popular effect in Macro photography.

(15) It would be beneficial to have a Macro camera in mobile devices that captures Macro images from a larger lens-object distance (e.g. 5-15 cm) with larger object to image magnification (e.g. 1:1-15:1), and which has a high degree of optical Bokeh.

SUMMARY

(16) In various embodiments, there are provided folded digital cameras, comprising: a lens system with a lens with $N \geq 6$ lens elements L.sub.i and having an EFL and a TTL, wherein each lens element has a respective focal length f.sub.i and wherein a first lens element L.sub.1 faces an object side; an image sensor; and an OPFE for providing a folded optical path between an object and the lens, wherein the lens system has a focusing range that covers object-lens distances from infinity to a minimal object distance (MIOD), and wherein $MIOD/EFL < 20$.

(17) In various embodiments, there are provided folded digital cameras, comprising: a lens system with a lens with $N \geq 6$ lens elements L.sub.i, having an EFL and a TTL, wherein each lens element L.sub.i has a respective focal length f.sub.i and wherein a first lens element L.sub.1 faces an object side; an image sensor; and an OPFE for providing a folded optical path between an object and the lens, and wherein $Max\ CRA/FOV < 0.25$ when the camera is focused at infinity.

(18) In some embodiments, $MIOD/EFL$ may be smaller than 15, 13, 10, or even than 7.

(19) In some embodiments, the maximum field curvature MFC for any object within the focus range may be smaller than 50 μm .

(20) In some embodiments, a camera may have a f number smaller than 4, 3, or even smaller than 2.5

(21) In some embodiments, the lens elements may be divided into two lens element groups separated by a big gap greater than $TTL/8$, $TTL/7$, or even greater than $TTL/6$.

(22) In some embodiments, the lens elements may be divided into a first lens element group with an effective focal length EFL1 and a second lens element group with an effective focal length EFL2, wherein a ratio $EFL1/EFL2$ may not deviate from 1 by more than 20%, or even by more than 10%

(23) In some embodiments, a camera may further comprise a focusing mechanism for focusing the camera based on a voice coil motor.

(24) In some embodiments, the first lens element L1 and a second lens element L2 may be made of

a material with an Abbe number greater than 50.

(25) In some embodiments, TTL/EFL may be smaller than 1.5, 1.4, or even smaller than 1.3.

(26) In some embodiments, a ratio of the focal length $f_{\text{sub.N}}$ of the last lens element $L_{\text{sub.N}}$ and the TTL, $f_{\text{sub.N}}/\text{TTL}$ may be smaller than 1.0, 0.9, 0.8, 0.75, or even smaller than 0.7.

(27) In some embodiments, Max CRA/FOV may be smaller than 0.2, or even smaller than 0.15.

(28) In some embodiments, a maximum field curvature MFC for any object within the focus range may be smaller than 50 μm .

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting examples of embodiments disclosed herein are described below with reference to figures attached hereto that are listed following this paragraph. The drawings and descriptions are meant to illuminate and clarify embodiments disclosed herein, and should not be considered limiting in any way.

(2) FIG. 1A illustrates schematically a known digital folded camera in a perspective view;

(3) FIG. 1B illustrates schematically the camera of FIG. 1A in a side view;

(4) FIG. 1C shows schematically a known dual-camera including a digital camera and an upright camera in a perspective view;

(5) FIG. 2A shows schematically an optical lens system disclosed herein;

(6) FIG. 2B shows schematically another optical lens system disclosed herein;

(7) FIG. 2C shows schematically yet another optical lens system disclosed herein;

(8) FIG. 2D shows schematically yet another optical lens system disclosed herein, focused to infinity;

(9) FIG. 2E shows the lens system of FIG. 2D focused to 100 mm;

(10) FIG. 2F shows the lens system of FIG. 2D focused to 50 mm;

(11) FIG. 2G shows schematically an optical lens system disclosed herein with lens elements cut to a non-circular shape;

(12) FIG. 2H shows schematically another optical lens system disclosed herein with lens elements cut to a non-circular shape;

(13) FIG. 2I shows schematically yet another optical lens system disclosed herein with lens elements cut to a non-circular shape;

(14) FIG. 2J shows schematically yet another optical lens system disclosed herein with lens elements cut to a non-circular shape;

(15) FIG. 2K shows schematically yet another optical lens system disclosed herein, focused to infinity;

(16) FIG. 3A illustrates the orthogonal projections $IP_{\text{sub.orth},1}$, $IP_{\text{sub.orth},2}$ of two impact points $IP_{\text{sub.1}}$ and $IP_{\text{sub.2}}$ on a plane P that is orthogonal to the optical axis of the lens of the system in FIGS. 2A-2D;

(17) FIG. 3B illustrates the orthogonal projections $IP_{\text{sub.orth},3}$, $IP_{\text{sub.orth},4}$ of two impact points $IP_{\text{sub.3}}$ and $IP_{\text{sub.4}}$ on a plane P that is orthogonal to the optical axis of the lens of the system in FIGS. 2A-2D;

(18) FIG. 4 provides graphically a definition of clear height;

(19) FIG. 5 provides graphically a definition of clear aperture;

(20) FIG. 6 provides a graphical illustration of diameter $H_{\text{sub.Li}}$ of lens element $L_{\text{sub.i}}$;

(21) FIG. 7 shows an exploded view of lens elements illustrating lens element width $W_{\text{sub.L}}$ and height $H_{\text{sub.L}}$.

DETAILED DESCRIPTION

(22) In the following detailed description, numerous specific details are set forth in order to provide

a thorough understanding. However, it will be understood by those skilled in the art that the presently disclosed subject matter may be practiced without these specific details. In other instances, well-known methods have not been described in detail so as not to obscure the presently disclosed subject matter.

(23) FIGS. 2A-2J illustrate various lens systems disclosed herein. All lens systems shown in FIG. 2A-J can be included in a folded camera such as shown in FIGS. 1A and 1B.

(24) FIG. 2A shows schematically an optical lens system disclosed herein and numbered **200**. Lens system **200** comprises a lens **204**, an optical element **206** and an image sensor **208**. System **200** is shown with ray tracing. Optical element **206** may be for example an infra-red (IR) filter, and/or a glass image sensor dust cover. Optical rays (after their reflection by prism **202**) pass through lens **204** and form an image on image sensor **208**. FIG. 2A shows 3 fields with 3 rays for each: the upper marginal-ray, the lower marginal-ray and the chief-ray. In the example of FIG. 2A, the optical rays pass through optical element **206** before impinging on image sensor **208**. This is however not limiting, and in some examples, optical element **206** is not present. That is, the optical element may be optional in some designs.

(25) Lens **204** includes a plurality of N lens elements $L_{\text{sub}.i}$ where “ i ” is an integer between 1 and N . In lens system **200**, N is equal to six. This is however not limiting and a different number of lens elements can be used. According to some examples, N is equal to or greater than 5. For example, N can be equal to 5, 6, 7, 8, 9 or 10. $L_{\text{sub}.1}$ is the lens element closest to the object (prism) side and $L_{\text{sub}.N}$ is the lens element closest to the image side, i.e. the side where the image sensor is located. This order holds for all lenses and lens elements disclosed herein. Lens elements $L_{\text{sub}.i}$ can be used e.g. as lens elements of a camera similar to camera **100**. The N lens elements are axial symmetric along an optical (lens) axis **210**. Each lens element $L_{\text{sub}.i}$ comprises a respective front surface $S_{\text{sub}.2i-1}$ (the index “ $2i-1$ ” being the number of the front surface) and a respective rear surface $S_{\text{sub}.2i}$ (the index “ $2i$ ” being the number of the rear surface). This numbering convention is used throughout the description. Alternatively, as done throughout this description, lens surfaces are marked as “ $S_{\text{sub}.k}$ ”, with k running from 1 to $2N$. The front surface and the rear surface can be in some cases aspherical. This is however not limiting.

(26) As used herein, the term “front surface” of each lens element refers to the surface of a lens element located closer to the entrance of the camera (camera object side), and the term “rear surface” refers to the surface of a lens element located closer to the image sensor (camera image side).

(27) As explained below, a clear height value $CH(S_{\text{sub}.k})$ and a clear aperture value $CA(S_{\text{sub}.k})$ can be defined for each surface $S_{\text{sub}.k}$ (for $1 \leq k \leq 2N$). $CA(S_{\text{sub}.k})$ and $CH(S_{\text{sub}.k})$ define optical properties of each surface $S_{\text{sub}.k}$ of each lens element. The CH term is defined with reference to FIG. 4 and the CA term is defined with reference to FIG. 5.

(28) In addition a height $H_{\text{sub}.Li}$ (for $1 \leq i \leq N$) is defined for each lens element $L_{\text{sub}.i}$. $H_{\text{sub}.Li}$ corresponds, for each lens element $L_{\text{sub}.i}$, to the maximal height of lens element $L_{\text{sub}.i}$ measured along an axis perpendicular to the optical axis of the lens elements. For a given lens element, the respective $H_{\text{sub}.Li}$ is greater than, or equal to the CH and the CA of the front and rear surfaces of this given lens element. Typically, for an axial symmetric lens element, $H_{\text{sub}.Li}$ is the diameter of lens element $L_{\text{sub}.i}$ as seen in FIG. 6. Typically, for an axial symmetric lens element, $H_{\text{sub}.Li} = \max\{CA(S_{\text{sub}.2i-1}), CA(S_{\text{sub}.2i})\} + \text{a mechanical part size}$. In general, in lens design the mechanical part size is defined as not contributing to the optical properties of the lens.

Consequently, one defines two heights of a lens element $L_{\text{sub}.i}$, an optical height $H_{\text{sub}.opt,i}$ (corresponding to the CA value) of an optically active area **602** and a geometrical height of the lens $H_{\text{sub}.Li}$ of an entire lens area **604** which covers an optically active and an optically inactive area. The mechanical part and its properties are defined below. The mechanical part size contribution to $H_{\text{sub}.Li}$ is typically 200-1000 μm .

(29) In lens system **200**, some of the surfaces of the lens elements are represented as convex, and

some are represented as concave. The representation of FIG. 2A is however not limiting, and a different combination of convex and/or concave surfaces can be used, depending on various factors such as the application, the desired optical power, etc.

(30) As shown in FIGS. 3A, 3B and 4, each optical ray that passes through a surface S.sub.k impinges this surface on an impact point IP. Optical rays enter optical lens system 200 from surface S.sub.1 and pass through surfaces S.sub.2 to S.sub.2N. Some optical rays can impinge on any surface S.sub.k but cannot/will not reach image sensor 208. For a given surface S.sub.k, only optical rays that can form an image on image sensor 208 are considered. CH(S.sub.k) is defined as the distance between two closest possible parallel lines, see lines 400 and 402 in FIG. 4 located on a plane P orthogonal to the optical axis of the lens elements. In the representation of FIGS. 3A and 3B, plane P is parallel to plane X-Y and is orthogonal to optical axis 302 such that the orthogonal projection IP.sub.orth of all impact points IP on plane P is located between the two parallel lines.

(31) The definition of CH(S.sub.k) does not depend on the object currently imaged, since it refers to the optical rays that “can” form an image on the image sensor. Thus, even if the currently imaged object is located in a black background that does not produce light, the definition does not refer to this black background since it refers to any optical rays that “can” reach the image sensor to form an image (for example optical rays emitted by a background which would emit light, contrary to a black background).

(32) For example, FIG. 3A illustrates the orthogonal projections IP.sub.orth,1, IP.sub.orth,2 of two impact points IP.sub.1 and IP.sub.2 on plane P which is orthogonal to optical axis 302. By way of example, in the representation of FIG. 3A, surface S.sub.k is convex.

(33) FIG. 3B illustrates the orthogonal projections IP.sub.orth,3, IP.sub.orth,4 of two impact points IP.sub.3 and IP.sub.4 on plane P. By way of example, in the representation of FIG. 3B, surface S.sub.k is concave.

(34) In FIG. 4, the orthogonal projection IP.sub.orth of all impact points IP of a surface S.sub.k on plane P is located between parallel lines 400 and 402. CH(S.sub.k) is thus the distance between lines 400 and 402.

(35) Attention is drawn to FIG. 5. According to the presently disclosed subject matter, a clear aperture CA(S.sub.k) is defined for each given surface S.sub.k (for $1 \leq k \leq 2N$), as the diameter of a circle, wherein the circle is the smallest possible circle located in a plane P orthogonal to the optical axis 108 and encircling all orthogonal projections IP.sub.orth of all impact points on plane P. As mentioned above with respect to CH(S.sub.k), the definition of CA(S.sub.k) also does not depend on the object which is currently imaged. As shown in FIG. 5, the circumscribed orthogonal projection IP.sub.orth of all impact points IP on plane P is a circle 500. The diameter of circle 500 defines CA(S.sub.k).

(36) Detailed optical data and surface data are given in Tables 1-3 for the example of the lens elements in FIG. 2A. The values provided for these examples are purely illustrative, and according to other examples other values can be used.

(37) Surface types are defined in Table 1. The coefficients for the surfaces are defined in Table 2. The surface types are: a) Plano: flat surfaces, no curvature b) Q type 1 (QT1) surface sag formula:

$$(38) \quad z(r) = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2 r^2}} + D_{\text{con}}(u) \quad (\text{Eq. 1}) \quad D_{\text{con}}(u) = u^4 \cdot \text{Math} \cdot \prod_{n=0}^N A_n Q_n^{\text{con}}(u^2)$$

$$u = \frac{r}{r_{\text{norm}}}, x = u^2 \quad Q_0^{\text{con}}(x) = 1 \quad Q_1^{\text{con}} = -(5 - 6x) \quad Q_2^{\text{con}} = 15 - 14x(3 - 2x)$$

$$Q_3^{\text{con}} = -\{35 - 12x[14 - x(21 - 10x)]\} \quad Q_4^{\text{con}} = 70 - 3x\{168 - 5x[84 - 11x(8 - 3x)]\}$$

$$Q_5^{\text{con}} = -[126 - x(1260 - 11x\{420 - x[720 - 13x(45 - 14x)]\})]$$

where {z, r} are the standard cylindrical polar coordinates, c is the paraxial curvature of the surface, k is the conic parameter, r.sub.norm is generally one half of the surface's clear aperture, and A.sub.n are the polynomial coefficients shown in lens data tables. The Z axis is positive towards image. Values for CA are given as a clear aperture radius, i.e. CA/2. The reference

wavelength is 555.0 nm. Units are in mm except for refraction index (“Index”) and Abbe #. Each lens element L.sub.i has a respective focal length f.sub.i, given in Table 1. The FOV is given as half FOV (HFOV). The definitions for surface types, Z axis, CA values, reference wavelength, units, focal length and HFOV are valid for Tables 1-17.

(39) TABLE-US-00001 TABLE 1 Lens system 200 EFL = 15.0 mm, F number = 2.2, Half FOV = 13.25 deg. Aperture Curvature Radius Focal Surface # Comment Type Radius Thickness (D/2) Material Index Abbe # Length 1 A.S Plano Infinity -0.795 3.408 2 Lens 1 ASP 5.103 1.539 3.645 Plastic 1.54 55.91 35.74 3 6.170 1.607 3.606 4 Lens 2 ASP 3.492 1.559 2.944 Plastic 1.54 55.91 17.63 5 4.611 0.620 2.792 6 Lens 3 ASP -4.169 1.421 2.772 Plastic 1.67 19.44 -14.69 7 -8.176 0.493 2.475 8 Lens 4 ASP 3.021 1.701 2.469 Plastic 1.54 55.91 20.11 9 3.337 3.954 2.438 10 Lens 5 ASP -3.359 1.509 2.942 Plastic 1.67 19.44 -39.43 11 -4.540 0.040 3.543 12 Lens 6 ASP 12.089 1.704 3.920 Plastic 1.54 55.91 12.05 13 -13.762 1.368 3.896 14 Filter Plano Infinity 0.210 — Glass 1.52 64.17 15 Infinity 0.275 — 16 Image Plano Infinity — —

(40) TABLE-US-00002 TABLE 2 Aspheric Coefficients Surface # Conic A4 A6 A8 A10 2 0 -2.64E-03 -8.84E-07 2.34E-07 -5.05E-07 3 0 -9.49E-03 6.22E-04 -3.76E-05 7.07E-07 4 0 -1.37E-02 -2.75E-04 9.19E-05 -7.12E-06 5 0 -1.48E-02 9.78E-04 -8.57E-05 2.50E-06 6 0 2.37E-02 -2.66E-03 2.23E-04 -7.42E-06 7 0 1.36E-02 -1.29E-03 8.14E-05 3.27E-06 8 0 -1.13E-02 -7.97E-04 4.89E-05 -8.25E-06 9 0 -8.28E-03 2.76E-05 -1.17E-04 5.77E-06 10 0 1.01E-02 -2.86E-04 5.62E-05 -2.92E-06 11 0 4.11E-03 -4.26E-04 6.78E-05 -3.01E-06 12 0 -2.04E-03 -2.68E-04 3.25E-05 -7.07E-07 13 0 -1.26E-03 -7.29E-05 -2.48E-06 6.02E-07

(41) Table 3 provides details on the variation of the properties of lens system **200** with the object-lens distance. The object-lens distance is defined by the distance of the object to the 1.sup.st principal plane of the lens.

(42) Table 4 provides details on the maximum (image-side) CRAs of lens system **200**. The maximum CRA and Half FOV (HFOV) are given for several object-lens distances (“Object”). Data refers to a field of 3.5 mm, corresponding to an edge of the image sensor (i.e. upper end of sensor diagonal).

(43) For achieving small values of maximum CRA, the focal length f.sub.N of the last lens element L.sub.N is smaller than the lens' TTL. The TTL of lens **204** is 18 mm. For lens **204**, f.sub.6=12.05 mm and a ratio of f.sub.N/TTL=**0.67**.

(44) TABLE-US-00003 TABLE 3 Lens system 200 Variation of lens properties with object distance Object Distance BFL HFOV [mm] [mm] [deg] Magnification Infinity 1.853 13.25 0.000 1000 2.076 13.17 -0.015 500 2.299 13.08 -0.030 300 2.599 12.98 -0.050 250 2.750 12.92 -0.060 200 2.977 12.83 -0.076 150 3.357 12.67 -0.101 100 4.125 12.34 -0.153

(45) TABLE-US-00004 TABLE 4 Max CRA/FOV-Lens system 200 HFOV (Image Height Object Max (“IH”) = 3.5 mm) Max CRA/FOV [mm] CRA [deg] (IH = 3.5 mm) Infinity 4.96 13.29 0.19 1000 5.59 13.22 0.21 500 6.09 13.14 0.23 300 6.02 13.06 0.23 250 5.76 13.01 0.22 200 5.87 12.93 0.23 150 6.02 12.80 0.24 100 6.22 12.51 0.25

(46) The focusing range of lens system **200** is from infinity to 100 mm. The focusing range of a lens system is defined as all object-lens distances that can be focused to by means of a camera mechanism that controls the distance between lens and image sensor. That is, for each object located within the focus range, a focusing mechanism can set a particular lens-image sensor distance that results in maximum contrast for the object's image. Maximum contrast means that for lens-image sensor distances other than the particular lens-image sensor distance, the object's contrast will decrease. The minimal object distance (MIOD) is defined as the lower limit of the focusing range, i.e. the MIOD is the smallest object-lens distance that the lens system can focus to. For lens system **200**, the MIOD is 100 mm. Lens system **200** can focus continuously from infinity to **100** mm, i.e. any focus position between Infinity to 100 mm (as well as any magnification between 0 and -0.153) can be realized.

(47) For focusing lens **204**, all lens elements are moved together. For changing focus from infinity to 100 mm, a lens movement (“lens stroke”) of 2.272 mm is required. For moving the lens, an actuator as known in the art may be used, e.g. a voice coil motor (VCM). A Hall sensor-magnet geometry for large stroke linear position sensing which is required for VCMs supporting large strokes such as 2 mm or more is described in the U.S. Provisional Patent Application No. 63,059,200. At the MIOD, lens system **200** achieves a magnification of -0.153 , corresponding to an object-image ratio of ca. 6.5:1. The HFOV decreases from 13.25 degrees when focused to infinity to 12.34 degrees when focused to the MIOD.

(48) For any object within the focus range, lens system **200** has a maximum field curvature (MFC) smaller than $50\text{ }\mu\text{m}$. MFC may be defined as follows: when placing a lens such as lens **204** at a distance v from a flat image sensor such as image sensor **208**, image points at the optical axis will be in perfect focus, but image points off the optical axis will come into focus not at the image sensor, but at a distance v' smaller than v , wherein v' is less than MFC for all image points.

(49) A lens such as lens **204** can be divided into two lens groups, a first lens group (“focusing group” or “G1”) and a second lens group (“CRA correction group” or “G2”). In lens **204**, the focusing group includes lens elements L.sub.1, L.sub.2, L.sub.3 and L.sub.4. The CRA correction group includes L.sub.4 and L.sub.5. The focusing group and the CRA correction group are separated spatially from each other by a big gap (BG) of 3.954 mm. All lens elements of G1 together have an EFL1=14.71 mm. All lens elements of G2 together have an EFL2=13.55 mm.

(50) In another lens system embodiment **250** shown in FIG. 2G and for achieving a folded lens system with low $f/\#$ and low lens height at the same time, lens elements are cut to a non-circular shape (“D-cut”). The cut lens elements may be obtained by cutting the large lens elements of lens **204** to a height of e.g. 6 mm (in Y direction), so that the lens is now a cut lens **204-C**. That is, the lens elements L.sub.1 of lens **204** that have height $H_{\text{sub.Li}} > 6\text{ mm}$ (i.e. L.sub.1, L.sub.5 and L.sub.6) are cut to 6 mm. Lens elements are cut in the Y-direction. The cut is performed symmetrically, i.e. the cut top part (defined by larger Y values in FIG. 2A) is identical in size to the cut bottom (defined by smaller Y values in FIG. 2A). The cut lens elements in **204-C** have no circular symmetry like that of lens elements **204**; their width is larger than their height, i.e. $W_{\text{sub.Li}} > H_{\text{sub.Li}}$ (see example in FIG. 7). Lens elements L.sub.1 of lens **204** that have height $H_{\text{sub.Li}} \leq 6\text{ mm}$ are not changed. As to the cut, cut lens elements L.sub.1, L.sub.5 and L.sub.6 have a large CA but still low CH. This is beneficial as in a folded lens design in which a lens height $H_{\text{sub.L}}$ may determine the camera's height, wherein the camera height is limited in general by a height of the host device. A folded lens having large CA and low CH is beneficial for having a low $f/\#$ folded lens that is compatible with e.g. a smartphone's height constraints. The CA of lens elements L.sub.2, L.sub.3 and L.sub.4 may be oriented in any direction, e.g. in Y direction. As of the cut design, the CA of lens element L.sub.1, L.sub.5 and L.sub.6 is oriented in the X direction (not visible here).

(51) Light is lost at extreme rays by cutting the lens elements, but no light loss is expected for center rays. Light loss is given as percentage of rays-through at the image plane at an image sensor boundary having coordinates (X, Y)=(0, 2.1 mm), i.e. moving up from the optical axis by 2.1 mm: Infinity: 92.49% 1 m: 92.22% 0.5 m: 91.94% 20 cm: 91.01% 10 cm: 89.43%

(52) In other embodiments, only one or only two lens elements L.sub.1 may be cut, i.e. may have $W_{\text{sub.Li}} > H_{\text{sub.Li}}$. In yet other embodiments, more than three lens elements L.sub.1 may be cut, i.e. may have $W_{\text{sub.Li}} > H_{\text{sub.Li}}$. In yet other embodiments, all lens elements L.sub.1 may be cut, i.e. may have $W_{\text{sub.Li}} > H_{\text{sub.Li}}$. In yet other embodiments, a cut lens may be achieved by cutting the large lens elements of lens **204** to a height of e.g. 6.5 mm, 5 mm or 4 mm (in the Y direction), i.e. lens elements L.sub.i that have height $H_{\text{sub.Li}} > 6.5\text{ mm}$, 5 mm or 4.5 mm may be cut to 6.5 mm, 5 mm or 4.5 mm respectively.

(53) Attention is now drawn to FIG. 2B, which depicts schematically another optical lens system disclosed herein and numbered **220**. Lens system **220** comprises a lens **204'** with a plurality of lens

elements, optical element **206** (optional) and image sensor **208**. Ray tracing is provided as in FIG. 2A. Detailed optical data and surface data are given in Tables 5, 6, 7 and 8. Table 8 provides details on the maximum CRAs of lens system **220**. Notation and field are identical to Table 4.

(54) FIG. 2B shows 3 fields with 3 rays for each: the upper marginal-ray, the lower marginal-ray and the chief-ray.

(55) The focusing range of lens system **220** is from infinity to 100 mm, i.e. the MIOD is 100 mm.

(56) For focusing with lens **204'**, all lens elements are moved together. For changing focus from infinity to 100 mm, a lens stroke of 2.237 mm is required. For moving the lens, an actuator as known in the art may be used, e.g. a VCM. At the MIOD, lens system **210** achieves a magnification of -0.15 , corresponding to an object-image ratio of ca. 6.7:1. The HFOV decreases from 13.29 degrees when focused to infinity to 12.51 degrees when focused to the MIOD. Any focus position between infinity to 100 mm (as well as any magnification between 0 and -0.15) can be realized. For any object within the focus range, lens system **220** has a maximum field curvature (MFC) $< 50 \mu\text{m}$.

(57) Lens **204'** can be divided into two groups, a first focusing group which includes lens elements L.sub.1, L.sub.2, L.sub.3, L.sub.4 and L.sub.5 and a second CRA correction group which includes L.sub.6 and L.sub.7. The focusing group and the CRA correction group are separated spatially from each other by a big gap of 4.960 mm. All lens elements of G1 together have an EFL1=14.08 mm, all lens elements of G2 together have an EFL2=13.94 mm. The TTL of lens **204'** is 18 mm.

(58) FIG. 2H an embodiment numbered **260** of a lens system with a D-cut lens **204'-C** based on lens **204'**. As in the description of FIG. 2G, light loss is given as percentage of rays-through at coordinates (X, Y)=(0, 2.1 mm): Infinity: 92.33% 1 m: 92.07% 0.5 m: 91.75% 20 cm: 90.76% 10 cm: 89.11%

(59) D-cut lens **204'-C** is obtained by cutting the large lens elements of lens **204'** to a height of e.g. 6 mm (in the Y direction), i.e. the lens elements L.sub.i of lens **204'** that have height H.sub.Li > 6 mm (i.e. L.sub.1, L.sub.6 and L.sub.7) are cut to 6 mm. In other embodiments, a cut lens may be achieved by cutting the large lens elements of lens **204'** to a height of e.g. 6.5 mm, 5 mm or 4 mm (in the Y direction), i.e. the lens elements L.sub.i that have height H.sub.Li > 6.5 mm, 5 mm or 4.5 mm may be cut to 6.5 mm, 5 mm or 4.5 mm respectively. For details on cut lenses it is referred to the description of FIG. 2A.

(60) TABLE-US-00005 TABLE 5 Lens system 220 EFL = 15.0 mm, F number = 2.2, Half FOV = 13.3 deg. Aperture Curvature Radius Focal Surface # Comment Type Radius Thickness (D/2) Material Index Abbe # Length 1 A.S Plano Infinity -1.030 3.408 2 Lens 1 ASP 4.603 1.619 3.633 Plastic 1.54 55.91 119.97 3 4.335 1.382 3.441 4 Lens 2 ASP 3.577 1.700 2.930 Plastic 1.54 55.91 9.38 5 9.836 0.571 2.729 6 Lens 3 ASP -25.617 0.260 2.710 Plastic 1.67 19.44 -5.17 7 4.059 0.651 2.619 8 Lens 4 ASP -5.649 0.671 2.605 Plastic 1.54 55.91 17.54 9 -3.704 0.040 2.518 10 Lens 5 ASP 3.303 1.675 2.649 Plastic 1.67 19.44 13.22 11 4.172 4.960 2.471 12 Lens 6 ASP -3.581 1.519 3.048 Plastic 1.67 19.44 -35.15 13 -4.939 0.040 3.530 14 Lens 7 ASP 12.341 1.703 3.857 Plastic 1.54 55.91 11.86 15 -12.986 0.780 3.890 16 Filter Plano Infinity 0.210 — Glass 1.52 64.17 17 Infinity 0.220 — 18 Image Plano Infinity — —

(61) TABLE-US-00006 TABLE 6 Aspheric Coefficients Surface # Conic A4 A6 A8 A10 2 0 $-2.62\text{E}-03$ $9.87\text{E}-06$ $-3.09\text{E}-07$ $-4.05\text{E}-07$ 3 0 $-9.81\text{E}-03$ $4.21\text{E}-04$ $-2.87\text{E}-05$ $4.48\text{E}-07$ 4 0 $-7.71\text{E}-03$ $-4.01\text{E}-05$ $-4.65\text{E}-05$ $1.34\text{E}-06$ 5 0 $9.24\text{E}-03$ $-1.27\text{E}-03$ $-1.66\text{E}-05$ $2.67\text{E}-06$ 6 0 $-1.68\text{E}-03$ $7.18\text{E}-05$ $1.10\text{E}-04$ $-1.03\text{E}-05$ 7 0 $-2.43\text{E}-02$ $2.85\text{E}-03$ $-8.38\text{E}-05$ $-7.26\text{E}-06$ 8 0 $1.45\text{E}-02$ $6.75\text{E}-04$ $-3.20\text{E}-05$ $-4.50\text{E}-06$ 9 0 $1.54\text{E}-02$ $1.39\text{E}-03$ $-1.53\text{E}-04$ $1.25\text{E}-05$ 10 0 $-6.69\text{E}-03$ $-1.41\text{E}-04$ $2.93\text{E}-05$ $-4.92\text{E}-06$ 11 0 $-6.80\text{E}-03$ $1.38\text{E}-03$ $-1.76\text{E}-04$ $5.48\text{E}-06$ 12 0 $7.57\text{E}-03$ $1.80\text{E}-04$ $4.85\text{E}-06$ $-5.12\text{E}-07$ 13 0 $2.74\text{E}-03$ $-1.10\text{E}-05$ $2.79\text{E}-05$ $-1.63\text{E}-06$ 14 0 $-1.70\text{E}-03$ $-5.47\text{E}-05$ $1.09\text{E}-05$ $-2.45\text{E}-07$ 15 0 $-2.92\text{E}-04$ $-1.37\text{E}-04$ $5.95\text{E}-06$ $1.00\text{E}-07$

(62) TABLE-US-00007 TABLE 7 Lens system 220 Variation of lens properties with object distance

Object Distance BFL HFOV [mm] [deg] Magnification Infinity 1.210 13.29 0.00 1000 1.432
13.22 -0.02 500 1.655 13.14 -0.03 300 1.952 13.06 -0.05 250 2.101 13.01 -0.06 200 2.324 12.93
-0.08 150 2.698 12.80 -0.10 100 3.447 12.51 -0.15

(63) TABLE-US-00008 TABLE 8 CRA/FOV-Lens system 220 HFOV (Image Height Object Max
("IH") = 3.5 mm) Max CRA/FOV [mm] CRA [deg] (IH = 3.5 mm) Infinity 3.90 13.29 0.15 1000
4.54 13.22 0.17 500 5.07 13.14 0.19 300 4.65 13.06 0.18 250 4.74 13.01 0.18 200 4.87 12.93 0.19
150 5.04 12.80 0.20 100 5.28 12.51 0.21

(64) FIG. 2C depicts schematically another optical lens system disclosed herein and numbered **230**.
Lens system **230** comprises a lens **204''**, optical element **206** (optional) and image sensor **208**. Ray
tracing is provided as in FIG. 2A-2B. Detailed optical data and surface data are given in Tables 9,
10, 11 and 12. Table 12 provides details on the maximum CRAs of lens system **220**. Notation and
field are identical to Table 4.

(65) FIG. 2C shows 3 fields with 3 rays for each: the upper marginal-ray, the lower marginal-ray
and the chief-ray. The focusing range of lens system **230** is from infinity to 76.4 mm, i.e. the MIOD
is 76.4 mm. For focusing with lens **204''**, all lens elements are moved together. For changing focus
from Infinity to 100 mm, a lens stroke of 2.881 mm is required. For moving the lens, an actuator
(e.g. a VCM) as known in the art may be used. At the MIOD, lens system **230** achieves a
magnification of -0.195, corresponding to an object-image ratio of ca. 5.1:1. The HFOV decreases
from 13.29 degrees when focused to infinity to 12.52 degrees when focused to the MIOD. Any
focus position between Infinity to 76.4 mm (as well as any magnification between 0 and -0.195)
can be realized. For any object within the focus range, lens system 230 has a MFC<50 μ m.

(66) Lens **204''** can be divided into two groups, a first focusing group that includes L.sub.1,
L.sub.2, L.sub.3, L.sub.4 and L.sub.5 and a second CRA correction group that includes L.sub.6,
L.sub.7 and L.sub.8. The focusing group and the CRA correction group are separated spatially from
each other by a big gap of 3.839 mm. All lens elements of G1 together have an EFL1=13.50 mm,
all lens elements of G2 together have an EFL2=11.85 mm. The TTL of lens **204''** is 18 mm.

(67) FIG. 2I shows an embodiment numbered **270** of a lens system with a cut lens **204''**-C based on
lens **204''**. As in the description of FIG. 2G, light loss is given as percentage of rays-through at
coordinates (X, Y)=(0, 2.1 mm): Infinity: 93.21% 1 m: 92.99% 0.5 m: 92.74% 20 cm: 92.01% 7.64
cm: 85.82%

(68) **204''**-C is obtained by cutting the large lens elements of lens **204''** to a height of e.g. 6 mm (in
Y direction). The lens elements L.sub.i of lens **204''** that have height H.sub.Li>6 mm (i.e. L.sub.1,
L.sub.2, L.sub.6, L.sub.7 and L.sub.8) are cut to 6 mm. In other embodiments, a cut lens may be
achieved by cutting the large lens elements of lens **204''** to a height of e.g. 6.5 mm, 5 mm or 4 mm
(in the Y direction). For details on cut lenses it is referred to the description of FIG. 2A.

(69) TABLE-US-00009 TABLE 9 Lens system 230 EFL = 15.0 mm, F number = 2.2, HFOV =
13.29 deg. Aperture Curvature Radius Focal Surface # Comment Type Radius Thickness (D/2)
Material Index Abbe # Length 1 A.S Plano Infinity -1.160 3.409 2 Lens 1 ASP 5.187 1.335
3.629 Plastic 1.54 56.02 138.46 3 5.064 0.950 3.445 4 Lens 2 ASP 3.563 1.513 3.150 Plastic
1.54 55.91 8.07 5 15.708 0.609 2.903 6 Lens 3 ASP -39.685 0.617 2.873 Plastic 1.65 21.78
-4.50 7 3.201 0.303 2.583 8 Lens 4 ASP -11.948 0.675 2.580 Plastic 1.53 55.97 22.38 9
-6.098 0.040 2.440 10 Lens 5 ASP 3.209 1.680 2.444 Plastic 1.67 19.44 10.03 11 4.813 3.839
2.212 12 Lens 6 ASP -3.377 1.002 2.840 Plastic 1.65 21.78 -6.59 13 -17.249 0.040 3.403 14 Lens
7 ASP -15.688 1.652 3.496 Plastic 1.58 30.11 9.50 15 -4.267 0.040 3.690 16 Lens 8 ASP 12.159
1.701 4.123 Plastic 1.54 56.02 13.86 17 -18.946 1.592 4.063 18 Filter Plano Infinity 0.210 —
Glass 1.52 64.17 19 Infinity 0.202 — 20 Image Plano Infinity 0.000 —

(70) TABLE-US-00010 TABLE 10 Aspheric Coefficients Surface # Conic A4 A6 A8 A10 2 0
-1.26E-03 2.06E-05 -1.02E-06 -3.50E-07 3 0 -5.16E-03 3.31E-04 -2.35E-05 2.89E-07 4
0 -4.34E-03 2.48E-04 -4.84E-05 1.77E-06 5 0 8.67E-03 -8.16E-04 -1.20E-05 4.10E-06 6
0 -3.47E-04 2.29E-05 5.60E-05 -4.03E-06 7 0 -3.01E-02 2.92E-03 -7.26E-05 -6.05E-06

8 0 1.89E-02 2.99E-04 -6.15E-05 -8.51E-07 9 0 2.09E-02 2.15E-03 -3.43E-04 9.74E-06 10
0 -1.05E-02 -5.90E-04 1.64E-04 -1.37E-05 11 0 -5.32E-03 7.82E-04 -2.81E-05 7.36E-06 12
0 9.47E-04 5.64E-04 6.09E-05 -2.44E-06 13 0 -3.85E-03 1.06E-04 4.63E-05 -2.32E-06 14 0
3.81E-04 -3.62E-05 1.72E-05 -7.92E-07 15 0 3.94E-03 4.70E-05 -9.66E-06 5.03E-07 16 0
5.83E-05 -8.90E-05 5.22E-06 4.24E-08 17 0 -1.59E-03 1.52E-04 -1.02E-05 4.46E-07

(71) TABLE-US-00011 TABLE 11 Lens system 230 Variation of lens properties with object distance Object Distance BFL HFOV [mm] [mm] [deg] Magnification Infinity 2.004 13.29 0.000
1000 2.225 13.24 -0.015 500 2.447 13.11 -0.030 300 2.742 13.18 -0.050 250 2.889 13.12 -0.060
200 3.110 13.02 -0.075 150 3.477 12.92 -0.100 100 4.209 12.69 -0.149 76.4 4.885 12.52 -0.195

(72) TABLE-US-00012 TABLE 12 CRA/FOV-Lens system 230 HFOV Max CRA/FOV Object Max (Image height = (Image Height = [mm] CRA 3.5 mm) 3.5 mm) Infinity 2.57 13.29 0.10 1000
3.21 13.24 0.12 500 3.72 13.11 0.14 300 3.35 13.18 0.13 250 3.45 13.12 0.13 200 3.59 13.02 0.14
150 3.78 12.92 0.15 100 4.06 12.69 0.16 76 3.99 12.52 0.16

(73) Attention is now drawn to FIGS. 2D-2F, which depict schematically yet another optical lens system disclosed herein and numbered **240**. Lens system **240** comprises a lens **204'''**, optical element **206** (optional) and image sensor **208**. Ray tracing is provided as in FIG. 2A-2C. Detailed optical data and surface data are given in Tables 13, 14, 15, 16 and 17. Table 17 provides details on the Max CRAs of lens system **240**. Notation and field are identical to Table 4. FIGS. 2D-2F show three fields with 3 rays for each: the upper marginal-ray, the lower marginal-ray and the chief-ray.

(74) FIG. 2D shows lens system **240** focused to infinity ("Config. A"), FIG. 2E shows lens system **240** focused to 100 mm ("Config. B") and FIG. 2F shows lens system **240** focused to 50 mm ("Config. C"). The object-lens distance focused on is given by Surface 0 in Table 15. The focusing range of lens system **240** is from infinity to 50 mm, i.e. the MIOD is 50 mm.

(75) Lens **204'''** is divided into two groups that move relative to each other for focusing. A first lens group ("G1") includes lens elements L.sub.1 and L.sub.2, and a second lens group ("G2") includes L.sub.3, L.sub.4, L.sub.5 and L.sub.6. The big gap of between G1 and G2 decreases from 2.625 mm when focused to Infinity to 1.946 mm when focused to 100 mm, and to 1.303 mm when focused to 50 mm (see Surface #5 in table 15). For focusing, lens **204'''** also moves as one unit so that the BFL changes (see Surface #13 in Table 15).

(76) For moving the lens, an actuator (e.g. VCM) may be used, At the MIOD, lens system **240** achieves a magnification of -0.40, corresponding to an object-image ratio of ca. 2.5:1. The HFOV decreases from 13.1 degrees when focused to infinity to 9.3 degrees when focused to the MIOD (see Table 16). Any focus position between Infinity to 50 mm (as well as any magnification between 0 and -0.4) can be realized.

(77) For any object within the focus range, lens system **240** has a MFC<50 μ m. All lens elements of G1 together have an EFL1=38.61 mm, all lens elements of G2 together have an EFL2=15.36 mm. The TTL of lens **204'''** is 15.8 mm.

(78) FIG. 2J shows an embodiment numbered **280** of a lens system with a cut lens **204'''**-C based on lens **204'''**. As in the description of FIG. 2G, light loss is given as percentage of rays-through at coordinates (X, Y)=(0, 2.1 mm): Infinity: 94.58% 10 cm: 91.53% 5 cm: 91.02%

(79) Cut lens **204'''**-C is obtained by cutting the large lens elements of lens **204'''** to a height of 6 mm (in Y direction), i.e. the lens elements L.sub.i of lens **204'''** that have height H.sub.Li>6 mm (L.sub.1, L.sub.2, L.sub.5 and L.sub.6) are cut to 6 mm. In other embodiments, a cut lens may be achieved by cutting the large lens elements of lens **204'''** to a height of e.g. 6.5 mm, 5 mm or 4 mm (in the Y direction).

(80) TABLE-US-00013 TABLE 13 Lens system 240 EFL = 15.0 mm, F number = 2.2, HFOV = 13.1 deg. Aperture Curvature Radius Focal Surface # Comment Type Radius Thickness (D/2) Material Index Abbe # Length 1 A.S Plano Infinity -1.138 3.394 2 Lens 1 ASP 5.548 1.713 3.399 Plastic 1.53 55.66 10.11 3 -225.776 0.031 3.268 4 Lens 2 ASP 4.454 1.000 3.084 Plastic 1.64 23.52 -9.96 5 2.401 2.625 2.627 6 Lens 3 ASP 4.197 1.000 2.722 Plastic 1.53 55.66 15.25

7 7.888 4.910 2.661 8 Lens 4 ASP -6.881 0.435 2.717 Plastic 1.66 20.37 1841.21 9 -7.016
0.162 2.848 10 Lens 5 ASP 2.335 0.551 2.906 Plastic 1.53 55.66 -24.20 11 1.816 1.274 3.339 12
Lens 6 ASP -7.488 1.251 3.393 Plastic 1.53 55.66 20.05 13 -4.675 0.344 3.546 14 Filter Plano
Infinity 0.210 — Glass 1.52 64.17 15 Infinity 0.290 — 16 Image Plano Infinity —

(81) For details on cut lenses it is referred to the description of FIG. 2A.

(82) TABLE-US-00014 TABLE 14 Aspheric Coefficients Surface # Conic A4 A6 A8 A10 2
-4.21E-01 6.50E-04 -1.01E-05 1.56E-06 -5.69E-08 3 -2.24E+02 2.36E-03 -1.58E-04
5.38E-06 -7.57E-08 4 -4.52E+00 -5.19E-04 5.31E-05 -6.85E-06 3.53E-07 5 -2.24E+00
2.21E-03 1.61E-05 -1.54E-06 7.95E-07 6 -8.98E-03 -7.71E-05 -5.52E-05 5.85E-06
-8.17E-07 7 6.54E-01 6.97E-04 -2.34E-05 -6.13E-07 -5.58E-07 8 -6.59E+01 4.55E-03
-1.90E-03 2.62E-05 3.32E-06 9 -4.32E-01 1.47E-02 -3.66E-03 2.57E-04 -5.76E-06 10
-4.54E+00 -2.21E-02 -1.28E-03 3.91E-04 -2.22E-05 11 -1.73E+00 -3.64E-02 4.33E-03
-2.94E-04 6.72E-06 12 -2.40E+01 1.32E-03 3.07E-04 -3.67E-05 -4.21E-07 13 -1.13E+00
6.63E-04 -3.99E-04 6.48E-05 -3.13E-06

(83) TABLE-US-00015 TABLE 15 Lens system 240 Variation of surface thicknesses Surface #
Config. A Config. B Config. C 0 1.00E+06 100 50 5 2.625 1.946 1.303 13 0.344 2.941 6.318

(84) TABLE-US-00016 TABLE 16 Lens system 240 Config. # HFOV Magnification A 13.1 deg 0
B 11.3 deg -0.17 C 9.3 deg -0.40

(85) TABLE-US-00017 TABLE 17 CRA/FOV-Lens system 240 HFOV (Image Height Object Max
("IH") = 3.5 mm) Max CRA/FOV [mm] CRA [deg] (IH = 3.5 mm) Infinity 12.65 13.09 0.48 1000
12.51 11.64 0.54 50 11.44 9.92 0.58

(86) TABLE-US-00018 TABLE 18 EFL ratios between groups G1 and G2 Last surface EFL1 EFL2
of first (EFL of (EFL of Ratio Embodiment group G1) G2) EFL1/EFL2 200 9 14.71 13.55 1.09 220
11 14.08 13.94 1.01 230 11 13.50 11.85 1.14 240 5 38.61 15.36 2.51 290 11 14.1 12.3 1.15

Table 18 shows an overview on the EFLs of all lens elements of the G1 and G2 respectively as well
as ratios EFL1/EFL2 for lens system embodiments **200**, **220**, **230**, **240** and **290**.

(87) FIG. 2K depicts schematically another optical lens system disclosed herein and numbered **290**.
Lens system **290** comprises a lens **204'''** with a plurality of lens elements, optical element **206**
(optional) and image sensor **208**. Ray tracing is provided as in FIG. 2A. Detailed optical data,
surface data and further lens properties are given in Tables **19**, **20**, **21** and **22**.

(88) The focusing range of lens system **290** is from infinity to 52 mm (MIOD=52 mm).

(89) For focusing with lens **204'''**, all lens elements are moved together. For changing focus from
infinity to 52 mm, a lens stroke of 4.507 mm is required. At the MIOD, lens system **290** achieves a
magnification of -0.29, corresponding to an object-image ratio of ca. 3.4:1. The HFOV decreases
from 9.57 degrees when focused to infinity to 8.52 degrees when focused to the MIOD. Any focus
position between infinity to 52 mm (as well as any magnification between 0 and -0.29) can be
realized. For any object within the focus range, lens system **290** has a MFC<50 μ m.

(90) Lens **204'''** can be divided into two groups, a first focusing group that includes L.sub.1,
L.sub.2, L.sub.3, L.sub.4 and L.sub.5 and a second CRA correction group that includes L.sub.6,
L.sub.7 and L.sub.8. The focusing group and the CRA correction group are separated spatially from
each other by a big gap of 3.974 mm. All lens elements of G1 together have an EFL1=14.1 mm, all
lens elements of G2 together have an EFL2=12.3 mm. The TTL of lens **204'''** is 18.6 mm.

(91) Some embodiments may include a cut lens based on lens **204'''**. The cut lens may be achieved
by cutting the large lens elements of lens **204'''** to a height of e.g. 6.5 mm, 5 mm or 4 mm (in the Y
direction). For details on cut lenses, it is referred to the description of FIG. 2A.

(92) TABLE-US-00019 TABLE 19 Embodiment 290 F = 15.5 mm, F number = 2.58, HFOV =
13.29 deg. Aperture Curvature Radius Focal Surface # Comment Type Radius Thickness (D/2)
Material Index Abbe # Length 1 A.S Infinity -0.732 3.000 2 Lens 1 ASP 5.487 1.243 3.000
Plastic 1.54 56.02 178.01 3 5.351 0.942 3.000 4 Lens 2 ASP 3.687 1.573 3.000 Plastic 1.54
55.91 8.24 5 17.247 0.659 3.000 6 Lens 3 ASP -34.749 0.716 2.969 Plastic 1.65 21.78 -4.57 7

3.302 0.310 2.644 8 Lens 4 ASP -13.318 0.684 2.641 Plastic 1.53 55.97 24.85 9 -6.772 0.041
2.494 10 Lens 5 ASP 3.284 1.758 2.490 Plastic 1.67 19.44 9.94 11 5.040 3.974 2.228 12 Lens 6
ASP -3.602 1.073 2.897 Plastic 1.65 21.78 -7.06 13 -18.159 0.044 3.000 14 Lens 7 ASP -14.373
1.615 3.000 Plastic 1.58 30.11 10.02 15 -4.342 0.041 3.000 16 Lens 8 ASP 14.157 1.757 3.000
Plastic 1.54 56.02 14.47 17 -17.049 1.743 3.000 18 Filter Plano Infinity 0.210 — Glass 1.52 64.17
19 Infinity 0.212 — 20 Image Plano Infinity — —

(93) TABLE-US-00020 TABLE 20 Aspheric Coefficients Surface # Conic A4 A6 A8 A10 A12 2 0
-1.21E-03 -6.93E-05 3.32E-05 -6.30E-06 5.62E-07 3 0 -5.07E-03 3.57E-04 -1.87E-05
-2.46E-06 4.29E-07 4 0 -4.06E-03 5.18E-04 -1.20E-04 1.22E-05 -8.59E-07 5 0
7.39E-03 -8.92E-05 -2.83E-04 6.07E-05 -6.11E-06 6 0 -4.46E-04 3.79E-04 -1.33E-04
3.83E-05 -5.13E-06 7 0 -2.57E-02 1.10E-06 1.63E-03 -5.53E-04 9.45E-05 8 0
1.88E-02 -2.83E-03 1.86E-03 -5.87E-04 9.51E-05 9 0 1.91E-02 2.54E-03 -8.30E-04
2.32E-04 -5.19E-05 10 0 -9.94E-03 4.29E-04 -4.14E-04 1.81E-04 -3.75E-05 11 0
-4.66E-03 1.18E-03 -3.00E-04 1.12E-04 -2.25E-05 12 0 7.88E-04 -4.64E-04 4.63E-04
-8.04E-05 7.21E-06 13 0 -2.84E-03 -9.77E-04 3.76E-04 -4.70E-05 2.74E-06 14 0
1.30E-03 -9.09E-04 2.99E-04 -4.17E-05 2.82E-06 15 0 4.93E-03 -8.53E-04 2.09E-04
-2.46E-05 1.42E-06 16 0 1.54E-03 -1.01E-03 1.89E-04 -1.87E-05 1.06E-06 17 0
-6.52E-04 -1.55E-04 1.55E-05 5.19E-07 -1.41E-07 Surface # Conic A14 A16 2 0
-2.71E-08 5.34E-10 3 0 -2.83E-08 6.99E-10 4 0 4.33E-08 -1.18E-09 5 0 3.31E-07
-7.68E-09 6 0 3.20E-07 -7.80E-09 7 0 -8.39E-06 3.03E-07 8 0 -7.83E-06 2.58E-07
9 0 6.16E-06 -2.88E-07 10 0 3.69E-06 -1.41E-07 11 0 2.26E-06 -6.86E-08 12 0
-3.08E-07 4.71E-09 13 0 -6.12E-08 -6.65E-11 14 0 -8.10E-08 4.86E-10 15 0 -3.30E-08
9.41E-11 16 0 -3.20E-08 3.98E-10 17 0 7.77E-09 -1.35E-10

(94) TABLE-US-00021 TABLE 21 Embodiment 290 Variation of lens properties with object
distance Object Distance BFL HFOV [mm] [mm] [deg] Magnification Infinity 2.166 9.57 0.00
1033 2.395 9.56 -0.01 517 2.625 9.59 -0.03 310 2.931 9.60 -0.05 258 3.083 9.46 -0.06 207 3.312
9.42 -0.07 155 3.693 9.34 -0.10 103 4.453 9.15 -0.15 52 6.673 8.52 -0.29

(95) TABLE-US-00022 TABLE 22 CRA/FOV-Embodiment 290 HFOV Max. (Image Height
Object CRA (“IH”) = 2.8 mm) Max. CRA/FOV [mm] [deg] [deg] (IH = 2.8 mm) Infinity 0.65 9.57
0.03 1033 0.96 9.56 0.05 517 0.47 9.59 0.02 310 0.47 9.60 0.02 258 1.82 9.46 0.10 207 2.05 9.42
0.11 155 2.37 9.34 0.13 103 2.86 9.15 0.16 52 3.74 8.52 0.22

(96) According to some examples, at least part of the lens elements can have a shape (profile) in
cross-section (in plane X-Y, which is orthogonal to the optical lens system and which generally
coincides with the optical axis) that is not circular. In particular, as shown for example in FIG. 7 for
a lens barrel **700** carrying lens elements **702**, at least some of the lens elements can have a width
W.sub.Li (measured along axis X) which is greater than their height H.sub.Li (measured along axis
Y). Barrel **700** further comprises a barrel wall (or surrounding) **704**. The height H.sub.Li can
correspond to the total height of the lens element (including the mechanical part). In some
embodiments, a lens element in lens barrel **700** may have a symmetry about axis Y and/or about
axis X. A non-circular lens barrel such as **700** maybe shaped according to the cut lens elements of a
lens such as e.g. lens **204-C**. As shown in FIG. 7, the height of a lens barrel may be only slightly
higher than the lens element having the largest height in the lens. For example, a lens barrel may be
0-0.5 mm higher than the highest lens element. A lens barrel having an identical height as the
highest lens element is described for example in co-owned international patent application
PCT/IB2018/050988, which is incorporated herein by reference in its entirety.

(97) According to some examples, W.sub.Li is substantially greater than H.sub.Li (for example, by
at least a percentage which is equal or greater than 20%, these values being not limiting). In some
examples, W.sub.Li may be greater than H.sub.Li by a percentage of 20-70%. Consider lens
element La of folded lens **204'** as an example: W.sub.L8 is greater than H.sub.L8 by a percentage
of 32%.

(98) It is appreciated that certain features of the presently disclosed subject matter, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the presently disclosed subject matter, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable sub-combination.

(99) Furthermore, for the sake of clarity the term “substantially” is used herein to imply the possibility of variations in values within an acceptable range. According to one example, the term “substantially” used herein should be interpreted to imply possible variation of up to 10% over or under any specified value. According to another example, the term “substantially” used herein should be interpreted to imply possible variation of up to 5% over or under any specified value. According to a further example, the term “substantially” used herein should be interpreted to imply possible variation of up to 2.5% over or under any specified value.

(100) Unless otherwise stated, the use of the expression “and/or” between the last two members of a list of options for selection indicates that a selection of one or more of the listed options is appropriate and may be made.

(101) It should be understood that where the claims or specification refer to “a” or “an” element, such reference is not to be construed as there being only one of that element.

(102) All references mentioned in this specification are herein incorporated in their entirety by reference into the specification, to the same extent as if each individual reference was specifically and individually indicated to be incorporated herein by reference. In addition, citation or identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present disclosure.

Claims

1. A folded digital camera, comprising: a lens system with a lens and an image sensor, the lens having $N \geq 6$ lens elements L_i , wherein a first lens element L_1 faces an object, and an effective focal length (EFL); and an optical path folding element (OPFE) for providing a folded optical path between the object and the lens, wherein the lens system has a focusing range that covers object-lens distances from infinity to a minimal object distance (MIOD), wherein $MIOD/EFL < 20$, wherein the lens is divided into two lens groups, wherein the lens elements within each lens group do not move with respect to each other, wherein the two lens groups move with respect to each other for focusing to any object-lens distance within the focusing range, wherein the lens has a total track length (TTL) and wherein $TTL/EFL < 1.5$, and wherein for the focusing to any object-lens distance within the focusing range, each of the two lens groups also move with respect to the image sensor.

2. The folded digital camera of claim 1, wherein the two lens groups are separated by a gap having a gap thickness, and wherein for changing focus from infinity to an object-lens distance that fulfills $500 \text{ mm} \geq \text{object-lens distance} \geq 100 \text{ mm}$, the gap thickness changes by more than 0.5 mm and by less than 1 mm.

3. The folded digital camera of claim 2, wherein $10 \text{ mm} < EFL < 20 \text{ mm}$.

4. The folded digital camera of claim 1, wherein the two lens groups are separated by a gap having a gap thickness, and wherein for changing focus from infinity to an object-lens distance that fulfills $500 \text{ mm} \geq \text{object-lens distance} \geq 50 \text{ mm}$, the gap thickness changes by more than 1 mm and by less than 2 mm.

5. The folded digital camera of claim 4, wherein $10 \text{ mm} < EFL < 20 \text{ mm}$.

6. The folded digital camera of claim 1, wherein $MIOD/EFL < 15$.

7. The folded digital camera of claim 1, wherein $MIOD/EFL < 10$.

8. The folded digital camera of claim 1, wherein $MIOD < 500 \text{ mm}$ and wherein $10 \text{ mm} < EFL < 30 \text{ mm}$.

9. The folded digital camera of claim 1, wherein $MIOD < 300$ mm and wherein $10 \text{ mm} < EFL < 30$ mm.
10. The folded digital camera of claim 1, wherein $MIOD < 250$ mm and wherein $10 \text{ mm} < EFL < 30$ mm.
11. The folded digital camera of claim 1, wherein $MIOD < 150$ mm and wherein $12.5 \text{ mm} < EFL < 20$ mm.
12. The folded digital camera of claim 1, wherein the camera has an f number < 4 .
13. The folded digital camera of claim 1, wherein the camera has an f number < 3 .
14. The folded digital camera of claim 1, wherein the camera has an f number < 2.5 .
15. The folded digital camera of claim 1, wherein the camera has an f number < 2.25 .
16. The folded digital camera of claim 1, further comprising a voice coil motor based focusing mechanism for focusing the camera.
17. The folded digital camera of claim 1, further comprising an optical element, wherein the optical element is located between the lens and the image sensor.
18. The folded digital camera of claim 1, wherein the folded digital camera is included in a mobile device.
19. The folded digital camera of claim 18, wherein the mobile device is a smartphone.
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